

Physics 7501 (Autumn 2026)
Quantum Mechanics 1 (graduate core course)

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Abstract. In this elementary course of quantum mechanics, we will learn the mathematical basis of quantum mechanics (i.e. linear algebra), the postulates of quantum theory, and how to use quantum mechanics to understand microscopic quantum phenomena.

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0 Course Information

Instructor

Professor **Yuan-Ming Lu**

Contact Information

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- **Office Phone:** 614-292-3084
- **Office:** 2012 Physics Research Building (PRB)

Office Hours

Friday 2:00–3:00 pm in PRB 2015

Lecture Logistics

- **Location:** Scott Lab N050
- **Time:** Wed. Fri. 11:10 am – 12:30 pm
- **Schedule:** See Google Spreadsheet

Teaching Assistant & Guided Group Work

- **Graduate Teaching Assistant (GTA):** Mr. Hyungjun Jeon (jeon.258@osu.edu)
- **GTA Office Hours:** Thursday 3:00–4:00 pm in PRB 2015
- **Guided Group Work (GGW) Sessions:** Thursdays 9:30 am – 11:00 am in PRB 0118. For questions regarding GGW sessions, please contact Dr. Beatriz E Burrola Gabilondo (burrolagabilondo.1@osu.edu).

Prerequisites

The course will assume physics and mathematics knowledge at the advanced undergraduate level (e.g., basic quantum mechanics, linear algebra, and differential equations), which we will review as needed.

Lecture Notes

Lecture notes will be uploaded to Carmen as PDF files generated from LaTeX sources at least 4 days before each lecture.

Main Textbooks (Recommended but Not Required)

- *Modern Quantum Mechanics* by J.J. Sakurai and Jim Napolitano (3rd Edition, Cambridge University Press, 2020)
- *Quantum Mechanics* (Vol. 3, Lectures on Theoretical Physics) by David Tong (Cambridge University Press, 2025). Problem sets available at <https://www.damtp.cam.ac.uk/user/tong/books>
- *Principles of Quantum Mechanics* by Ramamurti Shankar (2nd Edition, Plenum Press, 2011)
- *Quantum Computation and Quantum Information* by Michael Nielsen and Isaac Chuang (Cambridge University Press, 2011)

Other References

- *Lectures on Quantum Mechanics and on Topics in Quantum Mechanics*, David Tong, University of Cambridge
- *Quantum Mechanics: Non-Relativistic Theory* by Lev Landau and L. Lifshitz (2nd Edition, Pergamon Press)
- *Lectures on Quantum Mechanics* by Steven Weinberg (Cambridge University Press, 2013)
- *Quantum theory: a foundational approach* by Charis Anastopoulos (Cambridge University Press, 2024)
- *Quantum Mechanics* by Arjun Berera and Luigi Del Debbio (Cambridge University Press, 2021)

Reading Assignments

For each lecture, you will be expected to complete reading assignments before class. These will be announced on Carmen. On days when reading assignments are discussed in class, you must answer questions about the reading using Carmen "Quizzes" by 10:00 am the day of class. Your answers will not be graded for correctness; full credit is given for submitting on time. Late submissions will not be counted.

Homework

Homework will be assigned weekly on Carmen (except during holidays or exams). Submissions are due on Carmen by 11:59 pm on Fridays (or occasionally Wednesdays).

HW Quizzes

To verify individual effort in solving homework problems, a 10-minute quiz will take place during the last 10 minutes (12:20–12:30 pm) of lecture on homework due dates. Handwritten answers will be scanned and uploaded to Carmen/Gradescope.

Grading Scheme

All homework, quizzes, and exams will be graded by the GTA. Refer to Table 1 for the grading allocation.

Table 1: Grading basis for the course.

Item	Time	Assignments	Percentage
Reading Quizzes	10:00 am on lecture day	23	6%
Homework	Friday at 11:59 pm	12	18%
HW Quizzes	Last 10 mins of Friday lecture	12	18%
Mid Term Exam	Oct. 9 (Fri.), 11:30 am–1:30 pm	1	29%
Final Exam	Dec. 17 (Thu.), 12:00–2:00 pm	1	29%

Exams

- **Mid Term Exam:** In-person exam scheduled for 11:30 am–1:30 pm on Friday, Oct. 9 in PRB 1080 (Smith Seminar Room).
- **Final Exam:** In-person exam scheduled for 12:00–2:00 pm on Thursday, Dec. 17 in PRB 1080 (Smith Seminar Room).

Course Topics

- Linear algebra: mathematics of quantum mechanics
- Postulates of quantum mechanics

- Quantum entanglement
- Schrödinger equation and quantum dynamics
- Heisenberg's uncertainty principle
- One-dimensional potentials
- The harmonic oscillator and coherent states
- Squeezed states and quantum optics
- Symmetry and conservation laws
- Rotational symmetry and angular momentum
- Central potentials and the hydrogen atom
- Variational methods and WKB approximation
- Perturbation theory: time-independent and time-dependent
- Scattering theory

Shared Values

Please note The Ohio State University Shared Values.

Academic Misconduct

Academic integrity is essential to maintaining an environment that fosters excellence in teaching, research, and other educational activities. The Ohio State University and the Committee on Academic Misconduct (COAM, <https://oaa.osu.edu/resources/policies-and-procedures/committee-academic-misconduct>) expect that all students understand the University's Code of Student Conduct (<https://trustees.osu.edu/bylaws-and-rules/code>). Any activity that compromises academic integrity or subverts the educational process will be reported to COAM.

Artificial Intelligence and Academic Integrity

Generative AI tools are permitted as learning aids to support your understanding (*e.g.*, asking for alternative explanations or math reviews). However, every calculation, derivation, and explanation submitted must reflect your own understanding. You may not copy or lightly modify AI-generated solutions. AI tools are strictly prohibited during timed assessments, exams, and quizzes. When AI is permitted and used for homework, include a brief acknowledgment detailing the tool and how you verified its output.

Religious Accommodations

Ohio State accommodates students' religious beliefs and practices. Requests for accommodations must be communicated in writing to the instructor within the first 14 days of the semester. See the official policy at OAA Religious Holidays Policy.

Disability Statement & Illness

Students requesting disability accommodations should register with Student Life Disability Services (<https://slds.osu.edu/>) and notify the instructor as early as possible. If you are ill and need to miss class, notify the instructor immediately.

Intellectual Diversity

Ohio State is committed to open inquiry and intellectual diversity. Classroom discussions aim to foster critical thinking on course material without promoting any single point of view.

Grievances and Problem Solving

To resolve grievances concerning grades or academic practices, contact the instructor first, followed by the department chairperson, college dean, and provost (Faculty Rule 3335-8-23).

Creating an Environment Free from Harassment and Discrimination

The Ohio State University does not tolerate harassment, discrimination, or sexual misconduct. Reports can be submitted to the Civil Rights Compliance Office (CRCO) at <http://civilrights.osu.edu/>, by phone at 614-247-5838, or via email at civilrights@osu.edu.

1 Postulates and Mathematics of Quantum Mechanics

1.1 Key experiments that established quantum mechanics (8/26)

- Refer to Weinberg Chap 1. For further reading, try Anastopoulos Chap 2, Shankar 3.

Quantum mechanics was first motivated by a series of experiments that cannot be explained by classical mechanics and statistical physics. These exceptions to classical physics happen in the microscopic world, when the scale of physical quantities, such as energy, angular momentum and electric charge becomes comparable to certain characteristic scales.

The quantization of energy was observed in both light and atomic spectra, which motivated Planck, Einstein and others to propose a discretized (“quantized”) energy levels for photons and atoms. Bohr was able to explain the Rydberg formula[1] of hydrogen atom spectral lines under the assumption of a quantized angular momentum for electrons, which was later observed in the Stern-Gerlach experiment[2]. After de Broglie first proposed the wave nature of microscopic particles such as electrons, Davisson and Germer[3] were able to directly observe it in the diffraction pattern of electrons scattered by a nickel crystal. These experiments call for a new theory framework that defeats the intuition of classical physics.

1.1.1 Energy quantization of light in terms of photons

Young’s double-slit experiment clearly demonstrated the wave nature of light, whose energy should change continuously in classical electromagnetism. However, this point of view is seriously challenged by black-body radiation.

1.1.1.1 Black-body radiation and Planck’s formula

Consider light in a cubic box (a simplified model of a black body) of volume L^3 . A standing wave in the box satisfies the following condition for its wavevector $\vec{k} = (k_x, k_y, k_z)$:

$$\lambda_i = \frac{2\pi}{k_i} = \frac{2L}{n_i}, \quad n_i = 1, 2, 3, \dots, \quad i = x, y, z \quad (1.1)$$

Counting the number of standing wave modes in the box in terms of frequency $\nu = ck/2\pi$

$$\int dk_x dk_y dk_z = \frac{1}{8} 4\pi \int k^2 dk = \left(\frac{\pi}{L}\right)^3 \sum_{\{n_i\}} = \left(\frac{\pi}{L}\right)^3 N_{\text{modes}} \quad (1.2)$$

we have the density of states for light-wave modes per unit volume:

$$\frac{1}{L^3} \frac{dN}{d\nu} = \left(\frac{1}{\pi}\right)^3 \frac{\pi}{2} k^2 \frac{dk}{d\nu} = \frac{8\pi}{c^3} \nu^2 \quad (1.3)$$

Assuming the equipartition theorem in classical physics, i.e. each mode carries an averaged energy of $k_B T$, the black-body radiation energy per unit volume is

$$\frac{dE}{L^3 d\nu} = \frac{k_B T}{L^3} \frac{dN}{d\nu} = \frac{8\pi k_B T}{c^3} \nu^2 \quad (1.4)$$

This is known as the Rayleigh-Jeans formula and it correctly captures the long-wavelength (small ν) behavior of universal black-body radiation, first established by Kirchhoff around 1860. However, it diverges in the short wavelength limit, known as the ultraviolet catastrophe. Experimentally the Wien’s Distribution Law was observed at short wavelengths:

$$\frac{dE}{L^3 d\nu} \sim \nu^3 e^{-\alpha(T)\nu} \quad (1.5)$$

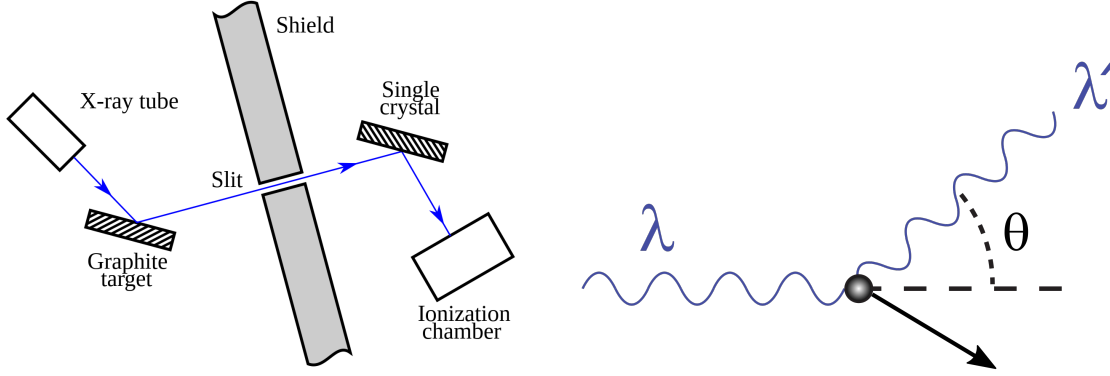


Figure 1.1: (Left) Schematic setup of Compton's experiment. (Right) The light-electron scattering process in the Compton experiment.

In order to reconcile long-wavelength Rayleigh-Jeans formula to short-wavelength Wien's law, Planck came up with his famous formula:

$$\frac{dE}{L^3 d\nu} = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{\alpha\nu} - 1}, \quad \alpha = \frac{h}{k_B T} \quad (1.6)$$

where the Planck constant is

$$h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s} \quad (1.7)$$

Planck's formula (1.6) indicates that the statistical-averaged energy of light at frequency ν is given by

$$U(\nu) = \frac{h\nu}{e^{h\nu/K_B T} - 1} = \frac{\sum_{n \in \mathbb{Z}_+} nh\nu e^{-nh\nu/K_B T}}{\sum_{n \in \mathbb{Z}_+} e^{-nh\nu/K_B T}} \quad (1.8)$$

under Planck's quantization hypothesis that the microscopic states of light have energy $nh\nu$ where $n \in \mathbb{Z}_+$ is a positive integer.

1.1.1.2 Einstein's "light quantum" and photoelectric effect

Motivated by Planck's quantization hypothesis, based on the assumption that light gives out energy in multiples of energy quantum $h\nu$, Einstein predicted the photoelectric effect

$$K_{\max} = eV = h\nu - \Phi \quad (1.9)$$

where $K_{\max}$ is the maximal kinetic energy of emitted photoelectrons when light of frequency ν shines on a metal, and Φ is the work function of the metal. V is minimal voltage that is required to stop photoelectron emission. This experiment was carried out by Millikan[4], who obtained Planck's constant

$$h = e \frac{dV}{d\nu} \quad (1.10)$$

1.1.1.3 Compton effect demonstrated quantization of light in terms of photons

Discovered by Arthur Compton in 1923[5], when the energy of the X ray photon (~ 17 keV) was significantly larger than the binding energy of the atomic electron, the electrons could be treated as being free after scattering. In this case, a *Compton shift* between the wavelengths

of injected light (λ) and the scattered light (λ') was observed to satisfy the following relation with the scattering angle θ (see Fig. 1.1):

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta) \quad (1.11)$$

where h is the Planck's constant.

The Compton shift can be explained by energy and momentum conservation, by assuming the photon energy of $E = h\nu = hc/\lambda$.

1.1.2 Quantization of atomic energy levels

1.1.2.1 Rydberg formula for the spectral lines in a hydrogen atom

After the discovery of dark lines in the spectrum of the sun at the beginning of 18th century, it was later realized that hot atomic gases absorb and emit light only at certain definite frequencies. Building on works of Balmer and others, Rydberg discovered the following rule[1] for the frequency of emitted light from hydrogen gases:

$$\nu = \frac{c}{\lambda} = cR_H \left(\frac{1}{n^2} - \frac{1}{m^2} \right), \quad m, n \in \mathbb{Z}_+. \quad (1.12)$$

This clearly demonstrated the discreteness of atomic energy levels $E_n \propto \frac{1}{n^2}$, further motivating Bohr's theory of the hydrogen atom.

1.1.2.2 Franck-Hertz experiment

In the Franck-Hertz experiment (1914) illustrated in Fig. 1.2, in a vacuum tube fitted with three electrodes (an electron-emitting, hot cathode; a metal mesh grid; and an anode), when a grid voltage is applied between the cathode and the anode, the electric current through the tube filled with thin mercury vapours, due to electrons that pass through the grid and reach the anode, is measured in the experiment. As shown in the middle figure, the current increases rapidly at low voltages, and then sharply drops almost back to zero at 4.9 V, with similar effects observed as 9.8 V and other multiples of 4.9 V. This can be understood by the quantized energy levels of mercury atoms. inside the single mercury droplet in the vacuum tube, as shown in Fig. 1.2.

1.1.3 Quantization of angular momentum

1.1.3.1 Bohr's explanation of hydrogen atom spectral lines

Rydberg's rule[1] indicates that the photon energy emitted by the hydrogen atom is given by

$$h\nu = E_m - E_n, \quad E_n = -\frac{hcR_H}{n^2}, \quad n \in \mathbb{Z}_+. \quad (1.13)$$

How to explain the $1/n^2$ quantization of hydrogen atom energy levels? According to classical mechanics, for an electron in the Coulomb force field of hydrogen nucleus, its speed v and radius r satisfies

$$\frac{mv^2}{r} = \frac{ke^2}{r^2} \implies E = \frac{1}{2}mv^2 - \frac{ke^2}{r} = -\frac{ke^2}{2r} \quad (1.14)$$

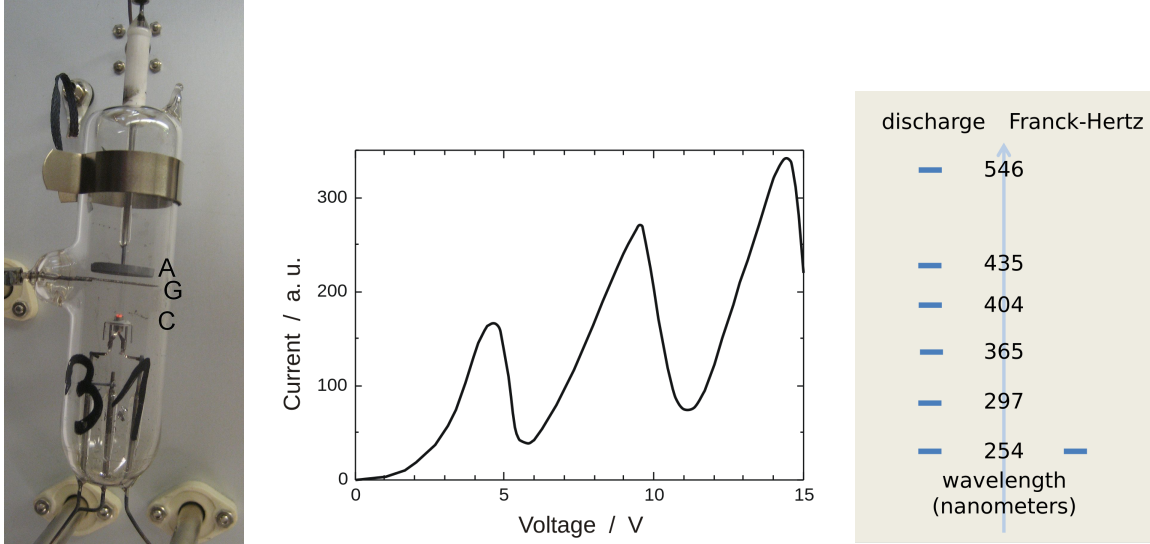


Figure 1.2: (Left) Photograph of a vacuum tube used for the Franck-Hertz experiment. (Middle) Anode current (arbitrary units) versus the grid voltage relative to the cathode. (Right) Wavelengths of light emitted by a mercury vapor discharge and by a Franck-Hertz tube in operation at 10 V. (Picture from Wikipedia)

where k is the Coulomb constant. To fit the n^2 dependence of the orbit radius r , Niels Bohr came up with the following quantization condition for angular momentum:

$$mvr = n\hbar \quad (1.15)$$

where $\hbar$ turns out to be $h/2\pi$. This quantization condition leads to the famous energy levels of electrons in a hydrogen atom:

$$E_n = -\frac{mk^2e^4}{2\hbar^2} \frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2} \quad (1.16)$$

1.1.3.2 Stern-Gerlach experiment demonstrated the quantization of angular momentum

In 1922 Stern and Gerlach performed an experiment[2] measuring the deflection of silver atoms by an inhomogeneous magnetic field, as illustrated in Fig. 1.3. Due to the interaction energy $U = -\vec{\mu} \cdot \vec{B}$ of a magnetic moment $\vec{\mu}$ under a magnetic field $\vec{B} = B_z \hat{z}$, a non-uniform magnetic field exerts a force $\vec{F} = -\vec{\nabla}U = \mu_z \vec{\nabla}B_z$ on the moment-carrying silver atoms, therefore deflecting the silver atom beams in the inhomogeneous magnetic field. As shown in Fig. 1.3, in contrast a continuous range of values of the magnetic moment as expected in classical mechanics, the experimental observations pointed to quantization of magnetic moment $\mu_z = g\mu_B S_z/\hbar$, where we defined μ_B as the Bohr magneton:

$$\mu_B = \frac{\hbar e}{2m_e} \quad (1.17)$$

$g \approx 2$ as the electron g -factor and

$$S_z = \pm \frac{\hbar}{2} \quad (1.18)$$

is the quantized spin angular momentum of a silver atom observed in Stern-Gerlach experiment.

This experiment demonstrated that in addition to energy level quantization, other physical observable such as spin angular momentum are also quantized in microscopic scales such as in

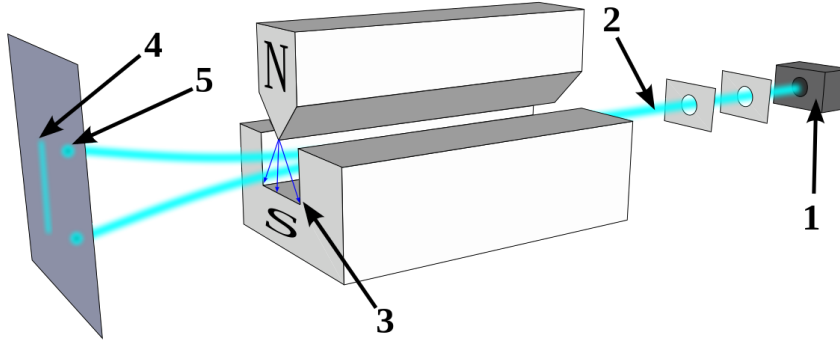


Figure 1.3: Schematic setup of Stern-Gerlach experiment: Silver atoms travelling through an inhomogeneous magnetic field, being deflected up or down depending on their spin. (1) furnace, (2) beam of silver atoms, (3) inhomogeneous magnetic field, (4) the classically expected result, (5) the observed result.

silver atoms. Moreover, we can consider sequential Stern-Gerlach experiments shown in Fig. 1.4, where silver atoms are deflected by multiple inhomogeneous magnetic fields in a sequence. Remarkably, unlike the classical case of light passing through polarizers, which filter out everything after two polarizers with perpendicular polarization axes, two quantum mechanical “beam splitters” with perpendicular magnetic field directions still allows two splitted beams to pass through, as illustrated in the middle and bottom parts of Fig. 1.4.

1.1.4 Davisson-Germer experiment demonstrated the wave nature of electrons

In 1924 de Broglie formally proposed the wave nature of electrons and other particles, in terms of their de Broglie wavelengths:

$$\lambda = \frac{h}{p} \quad (1.19)$$

where p is the momentum of the particle. If this proposal were true, it means that microscopic particles such as electrons should exhibit interference phenomena in the same way as acoustic and light waves.

In 1927 Davisson and Germer[3] performed an experiment to observe the diffraction pattern of electrons scattered by the surface of a crystal of nickel metal, as shown in Fig. 1.5. The highest intensity was observed at an angle $\theta = 50^\circ$ with a voltage of 54 V, giving the electrons a kinetic energy of 54 eV. To be compatible with the Bragg’s law

$$n\lambda = 2d \sin(90^\circ - \theta/2), \quad n \in \mathbb{Z}_+. \quad (1.20)$$

with the lattice spacing of nickel $d = 0.091$ nm, the $n = 1$ Bragg peak in (1.20) exactly corresponds to the de Broglie wavelength (1.19) of electrons. We will examine the experimental

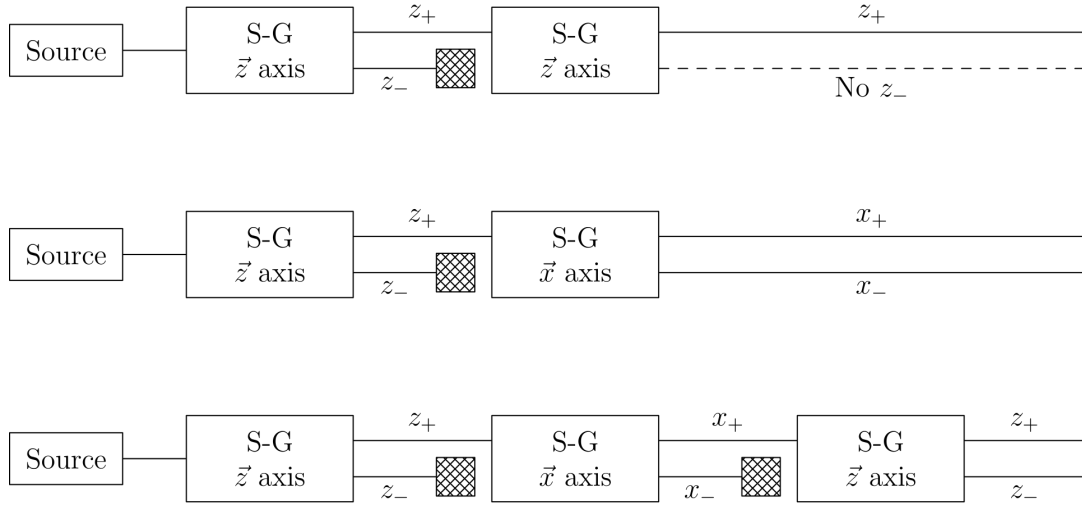


Figure 1.4: Sequential Stern-Gerlach experiments. (Picture from S&N)

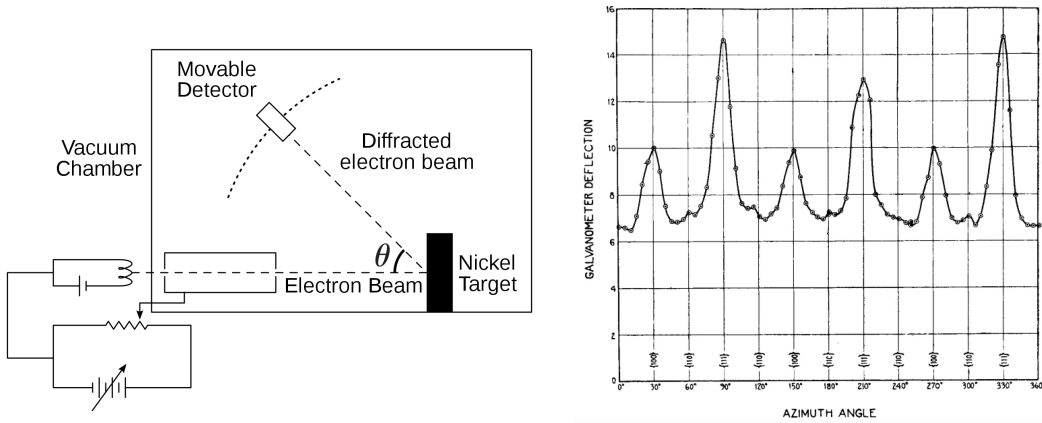


Figure 1.5: (Left) Schematic setup of Davisson-Germer experiment. (Right) Intensity of electron scattering vs. azimuth angle at 54 volts, co-latitude $\theta = 50^\circ$.

results in Fig. 1.5 more carefully in Homework 1.

1.2 Hilbert space and state vectors (8/28)

- Refer to N&C 2.1, Tong 3.1, Shankar 1.1-1.4

How to explain the above experimental observations? 4 postulates, derived (or more precisely, guessed) after a long process of trial and error, form the foundation for a theory framework that explains all observed experimental phenomena, known as quantum mechanics. Below we will introduce these postulates one by one, along the way of introducing linear algebra as the mathematics of quantum mechanics.

Postulate 1.1. *Associated to any isolated physical system is a complex vector space with inner product (i.e. a Hilbert space) known as the state space of the system. The system is completely described by its state vector, which is a unit vector in the system's state space.*

The key mathematical framework for quantum mechanics is linear algebra. The basic objects of linear algebra are vector spaces:

A *vector space* over a *scalar field* $\mathbb{F}$ is a set $\mathbb{V}$ of objects (called *vectors*) equipped two binary operations: a *vector addition* (“+”)

$$|v\rangle + |w\rangle \in \mathbb{V}, \quad \forall |v\rangle, |w\rangle \in \mathbb{V}. \quad (1.21)$$

and a *scalar multiplication*:

$$a|v\rangle \in \mathbb{V}, \quad \forall |v\rangle \in \mathbb{V}, a \in \mathbb{F}. \quad (1.22)$$

Mathematically, the set $\mathbb{V}$ forms an abelian group under vector addition operation and the scalar multiplication defines a ring homomorphism from the field $\mathbb{F}$ into the endomorphism ring of this abelian group. We will break down these terminology below.

It is worthwhile to define a group since we will study symmetry in quantum mechanics later. A group is a set $V = \{u, v, \dots\}$ equipped with a *closed* binary operation $+$: $V \times V \rightarrow V$

$$u + v = w \in V, \quad \forall u, v \in V. \quad (1.23)$$

where 3 group axioms are satisfied:

(1) *Identity element*:

$$\exists 0 \in V, \text{ s.t. } \forall u \in V, u + 0 = 0 + u = u. \quad (1.24)$$

(2) *Inverse*:

$$\forall u \in V, \exists (-u) \in V, \text{ s.t. } u + (-u) = 0. \quad (1.25)$$

(3) *Associativity*:

$$(u + v) + w = u + (v + w), \quad \forall u, v, w \in V. \quad (1.26)$$

An Abelian group is a group whose addition operation is commutative:

$$u + v = v + u, \quad \forall u, v \in V. \quad (1.27)$$

The scalar multiplication is another map $\mathbb{F} \times \mathbb{V} \rightarrow \mathbb{V}$ that is *associative*

$$a(b|v\rangle) = (ab)|v\rangle, \quad \forall a, b \in \mathbb{F}, |v\rangle \in \mathbb{V} \quad (1.28)$$

and *distributive in both the vectors and the scalars*:

$$a(|u\rangle + |v\rangle) = a|u\rangle + a|v\rangle, \quad (1.29)$$

$$(a + b)|v\rangle = a|v\rangle + b|v\rangle, \quad \forall a, b \in \mathbb{F}, |u\rangle, |v\rangle \in \mathbb{F}. \quad (1.30)$$

The above conditions satisfied by vector addition and scalar multiplication are summarized well in Chapter 1.1 of Shankar.

Of particular interests to quantum mechanics is a *Hilbert space*, which is a (n -dimensional) vector space $\mathbb{C}^n$ over the field $\mathbb{C}$ of complex numbers, equipped with a map $\mathbb{V} \times \mathbb{V} \rightarrow \mathbb{C}$ known as an *inner product*:

$$\langle u|v\rangle \in \mathbb{C}, \quad \forall |u\rangle, |v\rangle \in \mathbb{V}. \quad (1.31)$$

In the Dirac notation, $\langle u|$ is known as a *bra*, and $|v\rangle$ is known as a *ket*. The inner product is *skew-symmetric*

$$\langle u|v\rangle = \langle v|u\rangle^* \quad (1.32)$$

positive semidefinite

$$\langle v|v\rangle \geq 0, \quad \langle v|v\rangle = 0 \iff |v\rangle = 0. \quad (1.33)$$

and *linear in ket*

$$|u\rangle \left(\sum_i \lambda_i |v_i\rangle \right) = \sum_i \lambda_i \langle u|v_i\rangle \quad (1.34)$$

These properties are summarized in Chapter 1.2 of Shankar and Chapter 2.1.4 of N&C.

Note that an inner product also defines a norm for each vector in vector space $\mathbb{V}$:

$$||v\rangle| \equiv \sqrt{\langle v|v\rangle} \quad (1.35)$$

Two important properties of an inner product are

(1) **Cauchy-Schwarz inequality**:

$$\langle v|v\rangle \langle u|u\rangle \geq |\langle u|v\rangle|^2 \quad (1.36)$$

The equality holds if and only if $|v\rangle$ and $|w\rangle$ are linearly dependent.

(2) **Triangle inequality** for the norms of vectors:

$$||v\rangle + |w\rangle| \leq ||v\rangle| + ||w\rangle| \quad (1.37)$$

Triangle inequality can be derived from the Cauchy-Schwarz inequality.

Strictly speaking, in addition to being a vector space equipped with an inner product (known as an inner product space), a Hilbert space must also be *complete* in the sense that every Cauchy sequence $\{v^{(k)}\}$ of vectors in $\mathbb{V}$ has a limit that also lies in $\mathbb{V}$:

$$\lim_{k \rightarrow \infty} v^{(k)} \in \mathbb{V}, \quad \text{if } \forall \epsilon > 0, \quad \exists N \in \mathbb{Z}_+, \text{ s.t. } |v^{(m)} - v^{(n)}| < \epsilon, \quad \forall m, n > N. \quad (1.38)$$

For an inner product space of finite dimensions on any complete field $\mathbb{F}$, such as real and complex numbers $\mathbb{F} = \mathbb{R}, \mathbb{C}$, it is automatically a Hilbert space. Below are a few examples of Hilbert spaces, and inner product spaces that are not Hilbert spaces:

- n -dimensional Hilbert space on complex numbers: $\mathbb{V} \simeq \mathbb{C}^n$

- $n \times n$ complex matrices equipped with the Hilbert-Schmidt inner product forms a Hilbert space $L_{\mathbb{C}^n}$
- n -dimensional Hilbert space on real numbers: $\mathbb{V} \simeq \mathbb{R}^n$
- n -dimensional vector space $\mathbb{V} \simeq \mathbb{Q}^n$ on rational numbers is not a Hilbert space because it is not complete, since a Cauchy sequence of rational numbers can converge to an irrational number.
- The inner product space of continuous square-integrable complex functions on interval $[a, b]$ with the following L^2 inner product:

$$\langle f_1 | f_2 \rangle \equiv \int_a^b f_1^*(x) f_2(x) dx \quad (1.39)$$

is not a Hilbert space due to its incompleteness: a Cauchy sequence of continuous functions can be discontinuous. Its completion is the Hilbert space $L^2[a, b]$.

- For the same reason of incompleteness, the inner product space of polynomials on $[a, b]$ with the same L^2 inner product is not a Hilbert space, even though it's a subspace of $L^2[a, b]$.

A Hilbert space forms the *state space* described in Postulate 1.1. With the help of the inner product structure, we can always choose a complete set of orthonormal basis $\{|i\rangle\}$ that expands the full state space $\mathbb{V}$, such that

$$|v\rangle = \sum_i v_i |i\rangle, \quad \forall |v\rangle \in \mathbb{V}; \quad \langle i | j \rangle = \delta_{i,j} \quad (1.40)$$

where $\delta_{i,j}$ is the Kronecker delta function. Given a n -dimensional vector space V , one can always find a set of n linearly independent vectors $\{e_j | 1 \leq j \leq n\}$, known as a basis of vector space V , such that any vector $v \in V$ can be uniquely expressed as a linear superposition of the basis vectors: $v = \sum_j v_j e_j$ with $v_j \in \mathbb{F}$. Given the inner product structure in a Hilbert space, a set of complete orthonormal basis can be constructed following the Gram-Schmidt procedure, specified in Chapter 1.2 of Shankar and Chapter 2.1.4 of N&C. In such a complete orthonormal basis of Hilbert space $\mathbb{V} = \mathbb{C}^n$, a Dirac ket $|v\rangle$ can be represented by a column vector $\vec{v} = (v_1, \dots, v_n)^T$, a Dirac bra represented by a row vector, and the inner product reduces to the familiar form in $\mathbb{C}^n$:

$$\langle u | v \rangle = \sum_i u_i^* v_i = (u_1^*, \dots, u_n^*) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \quad (1.41)$$

Bibliography

- [1] J. R. Rydberg, The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science **29**, 331 (1890), <https://doi.org/10.1080/14786449008619945> .
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- [4] R. A. Millikan, Phys. Rev. **7**, 355 (1916).
- [5] A. H. Compton, Phys. Rev. **21**, 483 (1923).